Valentin Legras

Dear Hiring Team,

About myself

I am a Principal Data Scientist and scientific software engineer with 7+ years turning large-scale PK/PD and experimental datasets into decision-enabling tools across clinical and preclinical drug development. I combine hands-on machine learning and statistics with production-grade software engineering in R and Python, and I founded and led aNCA, an open-source R package now used by 120+ scientists. I am applying for the PKS Data Scientist & Scientific Software Engineer position in your Modeling & Simulation Data Science team, to keep building scalable, reusable analytical solutions that accelerate scientific decision making.

Why I fit the role

This role asks for a scientifically grounded data scientist who pairs analytical depth with real software engineering, and who delivers measurable impact through high-quality data products. That is exactly the profile I have built:

- **Production software engineering:** I established and enforced production standards for the aNCA package - version control, testing, documentation, code review, and CI/CD - taking it from concept to enterprise-wide adoption. Building robust, reusable, maintainable tools in a production setting is the core of what I do.
- **Machine learning and statistics:** I develop and evaluate ML models (Logistic Regression, Gradient Boosting, Random Forest) and apply explainable-AI techniques (feature importance, SHAP) to surface the drivers behind predictions, alongside a strong biostatistics foundation.
- **Scalable pipelines and applications:** I designed reproducible data pipelines and interactive R/Shiny and Spotfire applications that turn complex experimental data into insight, cutting manual effort by up to 90 percent and delivering more than 2M CHF in annual savings.
- **PK and preclinical data:** I bring a solid understanding of PK/PD and clearance concepts, and I extended the aNCA workflow to ingest and harmonize preclinical in vivo and in vitro PK data with protocol metadata - broadening it to pre-clinical, DMPK-adjacent use cases.
- **AI and automation:** I scale multi-agent coding workflows and LLM-based generative-AI tooling to automate complex analysis and development, boosting velocity while upholding engineering standards.

My mindset

Strategy and automation entered my life long before I knew the words for them. Competing in basketball across Europe from age 10 taught me that individual skill is never enough: it takes analyzing opponents, optimizing plays, and finding the most efficient way to win. Strategic video games sharpened the same instinct - reverse-engineering systems and automating processes to maximize outcomes with limited resources. That drive toward strategy, optimization, and automation is a core part of my identity; I still apply it outside work, building back-testing reproducible algorithms for S&P500 investment strategies (see GitHub), and I see AI as the tool that amplifies it at a scale that was not possible before.

Having led and mentored the aNCA team of 7 developers, I know my greatest impact comes from combining this optimization-first approach with the ability to align stakeholders and communicate clearly across multidisciplinary teams.

I would welcome the chance to discuss how my mix of data science, software engineering, and pharmacokinetic experience can contribute to your team.

Sincerely

Valentin Legras
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